Section A

1	(a)	force is proportional to the product of the masses and		
		inversely proportional to the square of the separation	M1	
		either point masses *or* separation >> size of masses	A1	[2]
	(b) (i)	gravitational force provides the centripetal force	B1	
		$mv^2/r = GMm/r^2$ and $E_K = \frac{1}{2}mv^2$	M1	
		hence $E_K = GMm/2r$	A0	[2]
	(ii) 1.	$\Delta E_K = \frac{1}{2} \times 4.00 \times 10^{14} \times 620 \times (\{7.30 \times 10^6\}^{-1} - \{7.34 \times 10^6\}^{-1})$	C1	
		$= 9.26 \times 10^7$ J (*ignore any sign in answer*)	A1	[2]
		(*allow* 1.0×10^8 J *if evidence that* E_K *evaluated separately for each* r)		
	2.	$\Delta E_P = 4.00 \times 10^{14} \times 620 \times (\{7.30 \times 10^6\}^{-1} - \{7.34 \times 10^6\}^{-1})$	C1	
		$= 1.85 \times 10^8$ J (*ignore any sign in answer*)	A1	[2]
		(*allow* 1.8 or 1.9×10^8 J)		
	(iii)	*either* $(7.30 \times 10^6)^{-1} - (7.34 \times 10^6)^{-1}$ *or* ΔE_K is positive / E_K increased	M1	
		speed has increased	A1	[2]
2	(a) (i)	sum of potential energy and kinetic energy of atoms / molecules / particles	M1	